MBA

MBA Fastrack 2025 (CAT + OMETs)

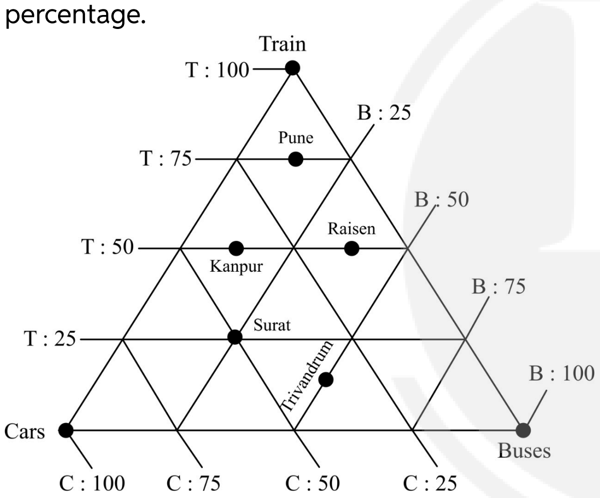
Data Interpretation & Logical Reasoning

DPP: 3

Miscellaneous Charts - 1

Directions (1-5) Read the following passage and answer the given questions.

A survey was conducted in five cities viz. Pune, Kanpur, Raisen, Surat, and Trivandrum, for the percentage of people using T (train), B (buses), and C (cars) as modes of transport. The number of persons surveyed in the cities of Pune, Kanpur, Raisen, Surat, and Trivandrum are 2000, 4000, 6000, 3000, and 8000 respectively. The following ternary diagram shows the data in percentage.



Q1 The average number of persons using trains for transportation in Pune, Kanpur, Raisen and Trivandrum is

(A) 1875 (B) 1850

(C) 1750 (D) 1650

Q2 The mode of transport used by the least number of persons in all the given cities is?

(A) Train

(B) Cars

(C) Buses

(D) Cannot be determined

Q3 Which of the follwoing statment(s) is/are TRUE?

(A) The number of people using Car in Surat is more than the number of people using Bus in Raisen.

(B) The number of people using Bus in Kanpur is less than the number of people using Car in Pune.

(C) Both A and B

(D) Neither A nor B

Q4 What is the ratio of the number of people using Trains to those using Buses across all cities?

(A) 33:30 (B) 31:29

(C) 33:31 (D) 35:32

Q5 What is the absolute difference between the number of people traveling by Train in Surat and those traveling by Car in Kanpur?

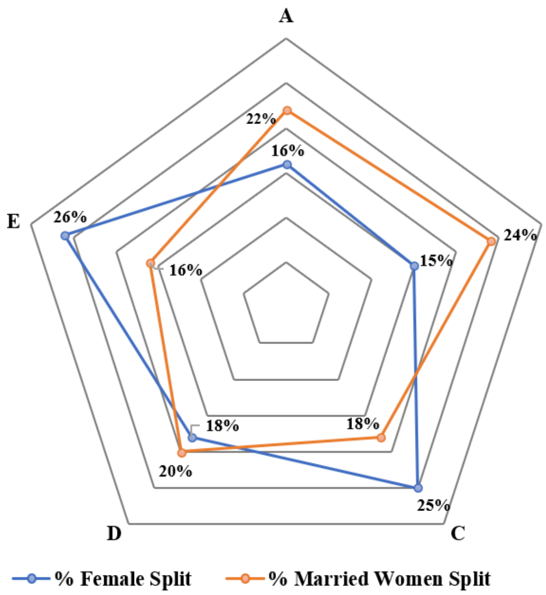
Directions (6-10) Read the following passage and answer the given questions.

Following chart represents the percentage break-ups of the number of females (married + unmarried) in different cities and the percentage break-ups of the number of married females.

Total number of females = 8400.

Total number of married females = 5200

Female & Married Female Splits



Q6 How many unmarried females are there in city B, D and E together?

(A) 1735 (B) 1836

(C) 1937 (D) 2038

Q7 What is the respective ratio of the number of married females in city E and number of unmarried females in city A?

(A) 99 : 37 (B) 48 : 13

(C) 104 : 25 (D) 57 : 22

Q8 Number of married females in city C and E together are approximately what percent of the number of unmarried females in C and E taken together?

(A) 70% (B) 75%

(C) 80% (D) 85%

Q9 If total population of city D is 3500 and the respective ratio of adult to non-adult male population is 3: 1, then how many adult males are there in city D if it is given that there are only males and females in city D?

(A) 1390 (B) 1491

(C) 1592 (D) 1693

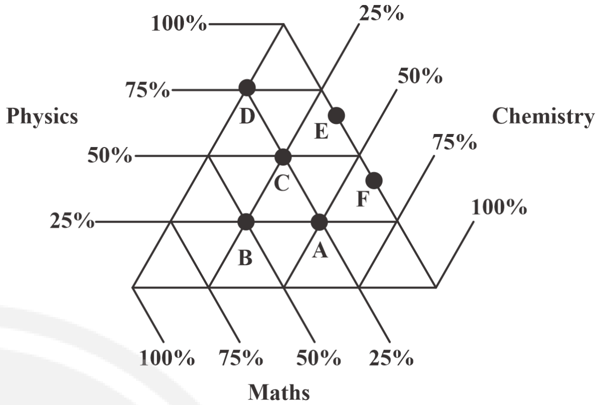
Q10 What is the difference between the female population of city B and married female population of city A and D together?

(A) 914 (B) 924

(C) 934 (D) 944

Directions (11-15) Read the following passage and answer the given questions.

The following ternary diagram shows the percentage of students in six cities – A, B, C, D, E and F whose favourite subject is Physics, Chemistry or Maths.



Q11 If the total number of students in city C is 120 and that in city D is 100, find the number of students whose favourite subject is Physics in C and D together .

(A) 100

(B) 115

(C) 135

(D) Cannot be determined

Q12 Find the difference between the number of students whose favourite subject is Maths from city B and the number of students whose favourite subject is Chemistry from city A.

(A) 12

(B) 15

(C) No difference

(D) Cannot be determined

Q13 If the number of students in city B is 96 and that in city C is 120, then the number of students in city B whose favourite subject is Physics is what percent of the number of students in city C whose favourite subject is Chemistry?

(A) 96% (B) 86%

(C) 80% (D) None of these

Q14 If the number of students in city A, B, C, D, E and F are 80, 96, 120, 100, 80 and 120 respectively, find the total number of students of these six cities whose favourite subject is Chemistry.

(A) 299 (B) 108
(C) 199 (D) 188

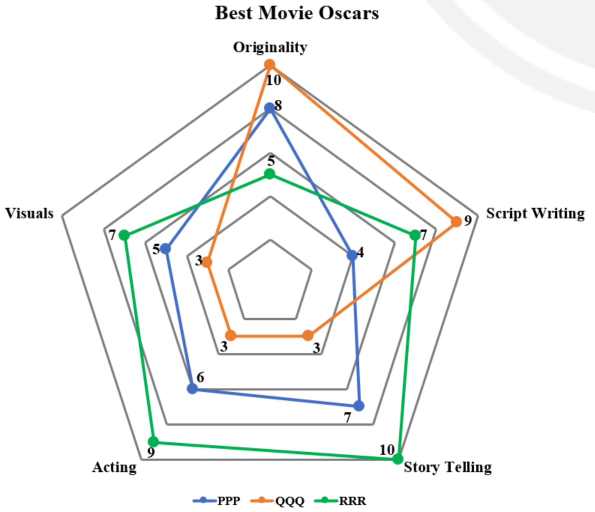
Q15 If the number of students in city A, B, C, D, E and F are 80, 96, 120, 100, 80 and 120 respectively, identify the city in which the highest number of students have their favourite subject as Physics.

(A) D (B) E
(C) A (D) C

Directions (16-20) Read the following passage and answer the given questions.

3 films were nominated for Oscars 23- PPP, QQQ & RRR. They were rated out of 10 across 5 segments – Originality, Script Writing, Story Telling, Acting, and Visuals. The film received the highest overall rating wins the Best Movies Award at Oscars.

Note : The overall rating of a film in any of the segments will be obtained by multiplying the weightage of that segment with the Individual Film rating



Q16 If the weightages of each of the 5 segments are 3, 4, 5, 6, and 2 respectively for Originality, Script Writing, Story Telling, Acting, & Visuals. The one whose overall rating is higher will win the Oscar and then find the winner(s) of the Oscar-

(A) PPP
(B) QQQ
(C) RRR
(D) Both PPP & QQQ

Q17 If the weightages of each of the 5 segments are 3, 4, 5, 6, and 2 respectively for Originality, Script Writing, Story Telling, Acting & Visuals, then find the ratio of the scores of QQQ & RRR?

(A) 11 : 16 (B) 15 : 23
(C) 11 : 15 (D) 16 : 23

Q18 If the weightages of each of the 5 segments are x, y, 7, 4, and 1 respectively for Originality, Script Writing, Story Telling, Acting & Visuals and it is also known that the net score of all the 3 films are the same, then find the value of (x - 2y).

(A) 10 (B) 9
(C) 8 (D) 7

Q19 If the film QQQ alone has won the Oscars then which of the below options can be the weightages of the 5 segments Originality, Script Writing, Story Telling, Acting & Visuals in the same order?

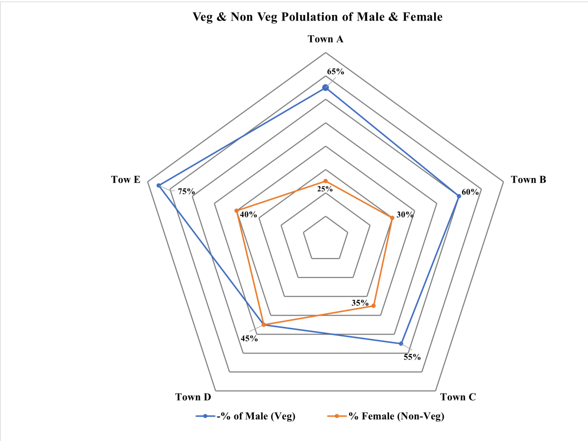
(A) 3,4,5,6,1 (B) 4,2,2,1,1
(C) 4,1,1,2,3 (D) 5,4,2,1,3

Q20 If the weightages across the 5 segments - Originality, Script Writing, Story Telling, Acting and Visuals are the first 5 prime numbers when arranged in the descending order, then who will win the Oscars?

(A) PPP
(B) QQQ
(C) RRR
(D) Both PPP & RRR

Directions (21-25) Read the following passage and answer the given questions.

Following chart represents the percentage of total number of vegetarian male population and percentage of non-vegetarian female population in 5 towns – A, B, C, D & E



The total population of Town A, B, C, D & E are 800, 750, 900, 1200 & 1000 respectively.

Q21 The ratio of male population to female population in town B is 3: 2 respectively and is reversed for town E. What is the difference between total vegetarian population of town B and total non-vegetarian population of town E?

(A) 120 (B) 130
(C) 140 (D) 150

Q22 Total vegetarian population of town A and D are 568 and 588 respectively. What is the respective ratio of total non-vegetarian male population of town A to the total non-vegetarian female population of town D?

(A) 11 : 19 (B) 14 : 27
(C) 8 : 13 (D) 10 : 17

Q23 The difference between total vegetarian female population and total non-vegetarian female population in town C is 108. If the ratio of male population to female population in town D are in the ratio 2: 3 respectively then total male population of town C is what percent of total male population of town D?

(A) 112.5% (B) 98.5%
(C) 106.5% (D) 92.5%

Q24 Total population of town E and F are in the ratio 2: 3 respectively and total male population and total female population in town F are in the ratio 8: 7 respectively. If total 865 people are non-vegetarian in town F out of which 4/5th are males, then find the total vegetarian female population of town F.

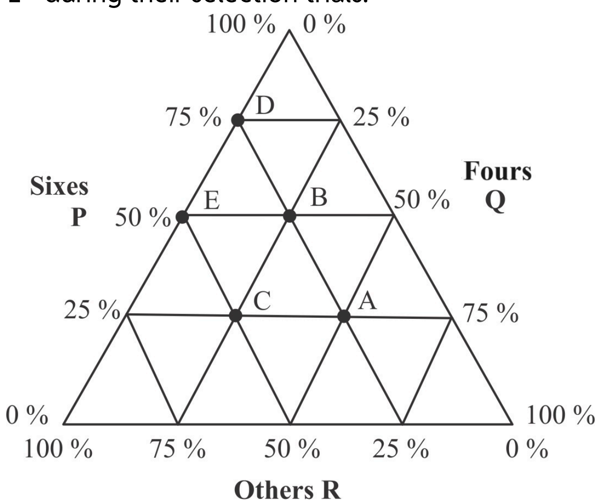
(A) 525 (B) 526
(C) 527 (D) 528

Q25 Total 104 males in town A are vegetarian who are 20% less than the total number of vegetarian males in town G. If total population of town B and G is same and the ratio of total vegetarian population and total non- vegetarian population in town G are in the ratio 4: 1 respectively, then what is the average of total vegetarian female population in town A and G taken together?

(A) 470 (B) 475
(C) 480 (D) 485

Directions (26-30) Read the following passage and answer the given questions.

The district level cricket board is selecting team members for its forthcoming season. The triangular graph given below shows the percentage distribution of runs under three categories - P (sixes) , Q (fours), and R (others) - that were scored by five batsmen - A, B, C, D, and E - during their selection trials.



Instructions to read the diagram: Coordinates of point A (P, Q, R) is (25%, 50%, 25%)

It is also known that:

(i) The total scores of two batsmen were 84 and 80. B had the highest total score.

(ii) The total runs scored by A was twice that of C.

(iii) The runs scored by B in sixes was 20% more than the runs scored by D in sixes.

(iv) The total score in 6's was PQ for A, while it was QP for E. (Note: PQ, and QP are two 2-digit numbers.)

(v) The runs scored in any single ball as "others" can be in 1s, 2s, or 3s only.

Q26 What was the difference between the total scores of B and E?

[Type '0' if the value cannot be uniquely determined]

Q27 What was the total number of runs that were scored by A, B, and C without hitting a six or a four?

(A) 108 (B) 84

(C) 72 (D) 80

Q28 How many sixes were hit by all five batsmen together?

[Type '0' if the value cannot be uniquely determined]

Q29 If each batsman faced 50 balls, then what is the maximum number of balls in which no runs were scored?

(A) 150 (B) 143

(C) 89 (D) 142

Q30 What are the lowest total runs scored by any player?

(A) 48 (B) 80

(C) 72 (D) None of these

Answer Key

Q1	(A)
Q2	(B)
Q3	(D)
Q4	(C)
Q5	750
Q6	(B)
Q7	(C)
Q8	(A)
Q9	(B)
Q10	(B)
Q11	(C)
Q12	(D)
Q13	(C)
Q14	(C)
Q15	(A)
Q16	(C)
Q17	(B)
Q18	(B)
Q19	(D)
Q20	(B)
Q21	(C)
Q22	(B)
Q23	(A)
Q24	(C)
Q25	(B)
Q26	60
Q27	(B)
Q28	35
Q29	(D)
Q30	(A)

Hints & Solutions

Note: scan the QR code to watch video solution

Q1. Text Solution:

Refering the ternary diagram:

Mode	Pune	Kanpur	Raisen	Surat	Trivandrum
Train	75%	50%	50%	25%	12.5%
Bus	12.5%	12.5%	37.5%	25%	50%
Car	12.5%	37.5%	12.5%	50%	37.5%
Total	2000	4000	6000	3000	8000

Calculating:

Mode	Pune	Kanpur	Raisen	Surat	Trivandrum	Total
Train	1500	2000	3000	750	1000	8250
Bus	250	500	2250	750	4000	7750
Car	250	1500	750	1500	3000	7000
Total	2000	4000	6000	3000	8000	23000

Required average = $\frac{1500+2000+3000+1000}{4}$

= $\frac{7500}{4}$

= 1875

Video Solution:

Q2. Text Solution:

Mode	Pune	Kanpur	Raisen	Surat	Trivandrum	Total
Train	1500	2000	3000	750	1000	8250
Bus	250	500	2250	750	4000	7750
Car	250	1500	750	1500	3000	7000
Total	2000	4000	6000	3000	8000	23000

Car is the mode of transport used by the least number of persons in all the given cities.

Video Solution:

Q3. Text Solution:

Mode	Pune	Kanpur	Raisen	Surat	Trivandrum	Total
Train	1500	2000	3000	750	1000	8250
Bus	250	500	2250	750	4000	7750
Car	250	1500	750	1500	3000	7000
Total	2000	4000	6000	3000	8000	23000

A. The number of people using Car in Surat is more than the number of people using Bus in Raisen: False

B. The number of people using Bus in Kanpur is less than the number of people using Car in Pune: False

Video Solution:

Q4. Text Solution:

Mode	Pune	Kanpur	Raisen	Surat	Trivandrum	Total
Train	1500	2000	3000	750	1000	8250
Bus	250	500	2250	750	4000	7750
Car	250	1500	750	1500	3000	7000
Total	2000	4000	6000	3000	8000	23000

The number of people using Trains across all cities = 8250

The number of people using Buses across all cities = 7750

Required ratio = 8250 : 7750

= 33 : 31

Video Solution:

Q5. Text Solution:

Mode	Pune	Kanpur	Raisen	Surat	Trivandrum	Total
Train	1500	2000	3000	750	1000	8250
Bus	250	500	2250	750	4000	7750
Car	250	1500	750	1500	3000	7000
Total	2000	4000	6000	3000	8000	23000

Train users in Surat = 750

Car users in Kanpur = 1500

Absolute Difference = 1500 - 750 = 750

Video Solution:

Q6. Text Solution:

For city A:

Number of females = 16% of 8400 = 1344

Number of married females = 22% of 5200 = 1144

Then, number of unmarried females = 1344 - 1144 = 200

Similarly,

City	Number of females		
	Total	Married	Unmarried
A	1344	1144	200
B	15% of 8400 = 1260.	24% of 5200 = 1248.	1260 – 1248 = 12
C	25% of 8400 = 2100.	18% of 5200 = 936.	2100 – 936 = 1164
D	18% of 8400 = 1512.	20% of 5200 = 1040.	1512 – 1040 = 472
E	26% of 8400 = 2184.	16% of 5200 = 832.	2184 – 832 = 1352

Number of unmarried females in city B, D and E together = 12 + 472 + 1352 = 1836.

Video Solution:

Q7. Text Solution:

For city A:

Number of females = 16% of 8400 = 1344

Number of married females = 22% of 5200 = 1144

Then, number of unmarried females = 1344 - 1144 = 200

Similarly,

City	Number of females		
	Total	Married	Unmarried
A	1344	1144	200
B	15% of 8400 = 1260.	24% of 5200 = 1248.	1260 – 1248 = 12
C	25% of 8400 = 2100.	18% of 5200 = 936.	2100 – 936 = 1164
D	18% of 8400 = 1512.	20% of 5200 = 1040.	1512 – 1040 = 472
E	26% of 8400 = 2184.	16% of 5200 = 832.	2184 – 832 = 1352

Number of married females in city E = 832

And, number of unmarried females in city A = 200

Therefore, ratio = 832 : 200 = 104: 25

Video Solution:

Q8. Text Solution:

For city A:

Number of females = 16% of 8400 = 1344

Number of married females = 22% of 5200 = 1144

Then, number of unmarried females = 1344 - 1144 = 200

Similarly,

City	Number of females		
	Total	Married	Unmarried
A	1344	1144	200
B	15% of 8400 = 1260.	24% of 5200 = 1248.	1260 – 1248 = 12
C	25% of 8400 = 2100.	18% of 5200 = 936.	2100 – 936 = 1164
D	18% of 8400 = 1512.	20% of 5200 = 1040.	1512 – 1040 = 472
E	26% of 8400 = 2184.	16% of 5200 = 832.	2184 – 832 = 1352

Number of married females in city C and E together $= 936 + 832 = 1768$

Number of unmarried females in city C and E together $= 1164 + 1352 = 2516$

Therefore, required percentage

$= \frac{1768}{2516} \times 100 = 70\%$ (approx.)

Video Solution:

Q9. Text Solution:

For city A:

Number of females = 16% of 8400 = 1344

Number of married females = 22% of 5200 = 1144

Then, number of unmarried females = 1344 - 1144 = 200

Similarly,

City	Number of females		
	Total	Married	Unmarried
A	1344	1144	200
B	15% of 8400 = 1260.	24% of 5200 = 1248.	1260 – 1248 = 12
C	25% of 8400 = 2100.	18% of 5200 = 936.	2100 – 936 = 1164
D	18% of 8400 = 1512.	20% of 5200 = 1040.	1512 – 1040 = 472
E	26% of 8400 = 2184.	16% of 5200 = 832.	2184 – 832 = 1352

In city D:

Total population $= 3500$

Female population $= 1512$

Then, male population $= 3500 - 1512 = 1988$

Therefore, adult male population

$= 1988 \times \frac{3}{4} = 1491$

Video Solution:

Q10. Text Solution:

For city A:

Number of females = 16% of 8400 = 1344

Number of married females = 22% of 5200 = 1144

Then, number of unmarried females = 1344 - 1144 = 200

Similarly,

City	Number of females		
	Total	Married	Unmarried
A	1344	1144	200
B	15% of 8400 = 1260.	24% of 5200 = 1248.	1260 – 1248 = 12
C	25% of 8400 = 2100.	18% of 5200 = 936.	2100 – 936 = 1164
D	18% of 8400 = 1512.	20% of 5200 = 1040.	1512 – 1040 = 472
E	26% of 8400 = 2184.	16% of 5200 = 832.	2184 – 832 = 1352

Female population of city B = 1260

And, married female population of city A and D together

= 1144 + 1040

= 2184

Therefore, difference = 2184 - 1260 = 924.

Video Solution:

Q11. Text Solution:

From the given Ternary diagram one can conclude the following table.

Subjects → ↓Cities	Physics	Chemistry	Maths
A	25 %	50 %	25 %
B	25 %	25 %	50 %
C	50 %	25 %	25 %
D	75 %	0 %	25 %
E	62.5 %	37.5 %	0 %
F	37.5 %	62.5 %	0 %

Number of students in city C whose favourite subject is Physics
= 50% of 120 = 60
Number of students in city D whose favourite subject is Physics
= 75% of 100 = 75
The total number of students whose favourite subject is Physics from both the cities = 60 + 75 = 135.

Video Solution:

Q12. Text Solution:

Number of students in city A whose favourite subject is Chemistry = 50% of total students in city A
Number of students in city B whose favourite subject is Maths = 50% of total students in city B
As, total number of students in city A and B is not given, we cannot calculate the required difference.

Video Solution:

Q13. Text Solution:

Number of students in city B whose favourite subject is Physics

$$= 25\% \text{ of } 96 = 24$$

Number of students in city C whose favourite subject is Chemistry

$$= 25\% \text{ of } 120 = 30.$$

Required percentage $= \frac{24}{30} \times 100 = 80\%.$

Video Solution:

Q14. Text Solution:

Number of students whose favourite subject is Chemistry in –
City A = 50% of 80 = 40.
City B = 25% of 96 = 24.
City C = 25% of 120 = 30.
City D = 0% of 100 = 0.
City E = 37.5% of 80 = 30.
City F = 62.5% of 120 = 75.
Total = 40 + 24 + 30 + 30 + 75 = 199.

Video Solution:

Q15. Text Solution:

Number of students whose favourite subject is Physics in –
City A = 25% of 80 = 20
City B = 25% of 96 = 24
City C = 50% of 120 = 60
City D = 75% of 100 = 75
City E = 62.5% of 80 = 50
City F = 37.5% of 120 = 45
Maximum 75 students are from city D.

Video Solution:

Q16. Text Solution:

Let's first simplify the data shared in the radar chart and present it in the tabular format-

Segments	Weightage	Individual Rating PPP	QQQ	RRR
Originality	3	8	10	5
Script Writing	4	4	9	7
Story Telling	5	7	3	10
Acting	6	6	3	9
Visuals	2	5	3	7

The overall rating of a film in any of the segments will be obtained if we multiply the weightage of that segment by the Individual Film rating. So, the overall rating of film data looks like below-

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	3	8	10	5	24	30	15
Script Writing	4	4	9	7	16	36	28
Story Telling	5	7	3	10	35	15	50
Acting	6	6	3	9	36	18	54
Visuals	2	5	3	7	10	6	14

The film with the highest rating wins the Oscars. So, the total rating looks like this -

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	3	8	10	5	24	30	15
Script Writing	4	4	9	7	16	36	28
Story Telling	5	7	3	10	35	15	50
Acting	6	6	3	9	36	18	54
Visuals	2	5	3	7	10	6	14
Overall					**121**	**105**	**161**

Thus, RRR is the winner of Oscars.

Video Solution:

Q17. Text Solution:

Let's first simplify the data shared in the radar chart and present it in the tabular format-

Segments	Weightage	Individual Rating PPP	QQQ	RRR
Originality	3	8	10	5
Script Writing	4	4	9	7
Story Telling	5	7	3	10
Acting	6	6	3	9
Visuals	2	5	3	7

The overall rating of a film in any of the segments will be obtained if we multiply the weightage of that segment with the Individual Film rating. So, the overall rating of film data looks like below-

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	3	8	10	5	24	30	15
Script Writing	4	4	9	7	16	36	28
Story Telling	5	7	3	10	35	15	50
Acting	6	6	3	9	36	18	54
Visuals	2	5	3	7	10	6	14

The film with the highest rating wins the Oscars. So, the total rating looks like this -

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	3	8	10	5	24	30	15
Script Writing	4	4	9	7	16	36	28
Story Telling	5	7	3	10	35	15	50
Acting	6	6	3	9	36	18	54
Visuals	2	5	3	7	10	6	14
Overall					**121**	**105**	**161**

Thus the ratio of the weighted scores of QQQ & RRR is 105 : 161 = 15 : 23.

Video Solution:

Q18. Text Solution:

Let's first simplify the data shared in the radar chart and present it in the tabular format-

Segments	Weightage	Individual Rating PPP	QQQ	RRR
Originality	3	8	10	5
Script Writing	4	4	9	7
Story Telling	5	7	3	10
Acting	6	6	3	9
Visuals	2	5	3	7

The overall rating of a film in any of the segments will be obtained if we multiply the weightage of that segment with the Individual Film rating. So, the overall rating of film data looks like below-

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	3	8	10	5	24	30	15
Script Writing	4	4	9	7	16	36	28
Story Telling	5	7	3	10	35	15	50
Acting	6	6	3	9	36	18	54
Visuals	2	5	3	7	10	6	14

The total rating looks like this :-

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	x	8	10	5	8x	10x	5x
Script Writing	y	4	9	7	4y	9y	7y
Story Telling	7	7	3	10	49	21	70
Acting	4	6	3	9	24	12	36
Visuals	1	5	3	7	5	3	7
Overall					8x + 4y + 78	10x + 9y + 36	5x + 7y + 113

So,

$(10x + 9y + 36) = (8x + 4y + 78)$

$= 2x + 5y = 42$ …..(I)

Also,

$(10x + 9y + 36) = (5x + 7y + 113)$

$= 5x + 2y = 77$ …..(II)

On solving I and II we get,11

$y = \frac{8}{3}$

$x = \frac{43}{3}$

So, $(x – 2y) = 9$

Video Solution:

Q19. Text Solution:

Let us try to find the overall scores corresponding to the weightages as mentioned in option A –

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	3	8	10	5	24	30	15
Script Writing	4	4	9	7	16	36	28
Story Telling	5	7	3	10	35	15	50
Acting	6	6	3	9	36	18	54
Visuals	1	5	3	7	5	3	7
Overall					116	102	154

By using this Weightage RRR will win the Oscar. Thus, clearly option A cannot be the answer.

Now let's look into the option B -

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	4	8	10	5	32	40	20
Script Writing	2	4	9	7	8	18	14
Story Telling	2	7	3	10	14	6	20
Acting	1	6	3	9	6	3	9
Visuals	1	5	3	7	5	3	7
Overall					65	70	70

Here, QQQ is not the only Oscars winner. Thus, this is not the right answer.

In option C we can create the below table –

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	4	8	10	5	32	40	20
Script Writing	1	4	9	7	4	9	7
Story Telling	1	7	3	10	7	3	10
Acting	2	6	3	9	12	6	18
Visuals	3	5	3	7	15	9	21
Overall					70	67	76

This is clearly not the answer.

Now, let's see what option D leads us to –

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	5	8	10	5	40	50	25
Script Writing	4	4	9	7	16	36	28
Story Telling	2	7	3	10	14	6	20
Acting	1	6	3	9	6	3	9
Visuals	3	5	3	7	15	9	21
Overall					91	104	103

Here we can clearly see that QQQ wins the Oscar.

Video Solution:

Q20. Text Solution:

The Weightage of Originality, Script Writing, Story Telling, Acting and Visuals is 11, 7, 5, 3, and 2 respectively. Now let's find the total ratings across all the segments as shown in the table below-

		Individual Rating			Weighted Rating		
Segments	Weightage	PPP	QQQ	RRR	PPP	QQQ	RRR
Originality	11	8	10	5	88	110	55
Script Writing	7	4	9	7	28	63	49
Story Telling	5	7	3	10	35	15	50
Acting	3	6	3	9	18	9	27
Visuals	2	5	3	7	10	6	14
Overall					179	203	195

As seen from the table, QQQ has the highest rating. So, QQQ is the winner.

Video Solution:

Q21. Text Solution:

Town	Total Population	Percentage of vegetarian male population	Percentage of non-vegetarian female population
A	800	65%	25%
B	750	60%	30%
C	900	55%	35%
D	1200	45%	45%
E	1000	75%	40%

In town B:

Total male population $= 750 \times \frac{3}{5} = 450$

Total female population $= 750 \times \frac{2}{5} = 300$

Then, total vegetarian population $= 60\%$ of $450 + (100 - 30)\%$ of $300 = 480$

In town E:

Total male population $= 1000 \times \frac{2}{5} = 400$

Total female population $= 1000 \times \frac{3}{5} = 600$

Then, total non-vegetarian population

$(100 - 75)\%$ of $400 + 40\%$ of $600 = 340$

Therefore, difference between total vegetarian population of town B and total non-vegetarian population of town $E = 480 - 340 = 140$.

Video Solution:

Q22. Text Solution:

Town	Total Population	Percentage of vegetarian male population	Percentage of non-vegetarian female population
A	800	65%	25%
B	750	60%	30%
C	900	55%	35%
D	1200	45%	45%
E	1000	75%	40%

In town A:

Let total male population = a

Total female population = 800 - a

Then, 568 = 65% of a + (100-25)% of (800 - a)

a = 320

Then, total non-vegetarian male population = (100 - 65) % of 320 = 112

In town D:

Let total male population = b

Total female population = 1200 - b

Then, 588 = 45% of b + (100 - 45) % of (1200 - b)

b = 720

Then, total non-vegetarian female population = 45% of (1200 - 720) = 216

Therefore, ratio = 112: 216 = 14: 27

Video Solution:

Q23. Text Solution:

Town	Total Population	Percentage of vegetarian male population	Percentage of non-vegetarian female population
A	800	65%	25%
B	750	60%	30%
C	900	55%	35%
D	1200	45%	45%
E	1000	75%	40%

Total female population of town

$C = 108 \times \frac{100}{65 - 35} = 360$

Then, total male population of town

$C = 900 - 360 = 540$

And, total male population of town

$D = 1200 \times \frac{2}{5} = 480$

Therefore, percentage $= \frac{540}{480} \times 100 = 112.5\%$

Video Solution:

Q24. Text Solution:

Town	Total Population	Percentage of vegetarian male population	Percentage of non-vegetarian female population
A	800	65%	25%
B	750	60%	30%
C	900	55%	35%
D	1200	45%	45%
E	1000	75%	40%

Total population of town $F = 1000 \times \frac{3}{2} = 1500$

Total female population of town

$F = 1500 \times \frac{7}{15} = 700$

Total non-vegetarian female population of town

$F = 865 \times \frac{1}{5} = 173$

Therefore, total vegetarian female population of town $F = 700 - 173 = 527$

Video Solution:

Q25. Text Solution:

Town	Total Population	Percentage of vegetarian male population	Percentage of non-vegetarian female population
A	800	65%	25%
B	750	60%	30%
C	900	55%	35%
D	1200	45%	45%
E	1000	75%	40%

Total vegetarian male population of town $A = 104$

Then, total non-vegetarian male population of town $A = 104 \times \frac{7}{13} = 56$

Then, total female population of town

$A = 800 - 104 - 56 = 640$

Now, total vegetarian female population of town

$A = (100 - 25)\%$ of $640 = 480$

Total vegetarian male population of town

$G = 104 \times \frac{100}{80} = 130$

Total population of town $G = 750$

Then, total vegetarian population of town

$G = 750 \times \frac{4}{5} = 600$

Now, total vegetarian female population of town

$G = 600 - 130 = 470$

Therefore, average of total vegetarian female population in town A and G taken together

$$= \frac{480 + 470}{2}$$

$$= 475$$

Video Solution:

Q26. Text Solution:

The percentage distribution of runs made from the triangular graph will be as follows:

Batsman	Sixes	Fours	Others
A	25%	50%	25%
B	50%	25%	25%

C	25%	25%	50%
D	75%	0%	25%
E	50%	0%	50%

From condition (i), The only person whose score can be 84 is E, as he scores 50% of the runs through sixes and the remaining 50% through others. (For any of the other 84 cannot be divided as integral multiples of 4 and 6 as per the given percentages.)

B cannot have 80 as his score as he is the highest scorer.

80 can be the score of D as it is divided into 75% and 25% as 60 and 20.

From condition (iv), The score of E through sixes was 42 = QP

So the score of A through sixes was PQ = 24.

Hence, A's total score was 96.

From condition (ii), C's total score was 48.

From condition (iii), The runs scored by B through sixes was 20% more than that of D:

D scored 60 runs through sixes, so B must have scored 60 x 1.2 = 72 runs through sixes.

Hence, B's total score was 144.

So we have the final table as follows.

Batsman	Sixes	Fours	Others	Total
A	24	48	24	96
B	72	36	36	144
C	12	12	24	48
D	60	0	20	80
E	42	0	42	84

Required difference = 144 – 84 = 60.

Video Solution:

Q27. Text Solution:

Batsman	Sixes	Fours	Others	Total
A	24	48	24	96
B	72	36	36	144
C	12	12	24	48
D	60	0	20	80
E	42	0	42	84

Required total = 24 + 36 + 24 = 84

Video Solution:

Q28. Text Solution:

Batsman	Sixes	Fours	Others	Total
A	24	48	24	96
B	72	36	36	144
C	12	12	24	48
D	60	0	20	80
E	42	0	42	84

Required number of sixes = $\frac{24+72+12+60+42}{6}$

= 35

Video Solution:

Q29. Text Solution:

Batsman	Sixes	Fours	Others	Total
A	24	48	24	96
B	72	36	36	144
C	12	12	24	48
D	60	0	20	80
E	42	0	42	84

Total balls faced = 250

The number of balls hit for sixes = $\frac{24+72+12+60+42}{6}$ = 35

The number of balls hit for fours = $\frac{48+36+12}{4}$ = 24

The total number of balls in which 6's and 4's were hit = 35 + 24 = 59

To find the maximum number of balls with no runs, we have to minimize the number of balls with runs scored.
Since the maximum number of other runs that can be scored in a ball is 3, we get the total number of balls in which runs were scored for each batsman as follows: A = 8; B = 12; C = 8; D = 6 + 1 = 7; E = 14;
Total = 8 + 12 + 8 + 7 + 14 = 49
Hence, the total number of balls in which runs were not scored = 250 – 59 – 49 = 142

Video Solution:

Q30. Text Solution:

Batsman	Sixes	Fours	Others	Total
A	24	48	24	96
B	72	36	36	144
C	12	12	24	48
D	60	0	20	80
E	42	0	42	84

C scored the lowest runs i.e. 48

Video Solution: